Ruihan Xu (Multy)

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RESEARCH INTERESTS

Spatial intelligence for embodied agents, with a focus on lifelong mapping and learning, 3D/4D scene representations, and robot planning and perception with uncertainty.

EDUCATION

Massachusetts Institute of Technology — **Cambridge, MA**
S.M.–Ph.D. Student, Aeronautics and Astronautics; Laboratory for Information & Decision Systems • GPA: 5.0/5.0 — Sep 2025 – Present

University of Michigan – Ann Arbor — **Ann Arbor, MI**
B.S. in Computer Science with Highest Honors (top 3%), Mathematics Minor • GPA: 4.0/4.0 — Dec 2024

PUBLICATIONS

* equal contribution

[1] **Ruihan Xu**, Jiae Yoon, Kaichen Zhou, Ue-Hwan Kim, Luca Carlone, Argos: Adapt Rich Geometric Priors for Generalizable Online Scene-Change-Detection, *arXiv preprint arXiv:2610.10181*, 2026, under review. [arXiv][Website]

[2] Hyungtae Lim, Nathan Hughes, Xihang Yu, **Ruihan Xu**, Yun Chang, Jingnan Shi, Rajat Talak, Luca Carlone, Hydra++: Real-Time Hierarchical 3D Scene Graph Construction With Object-Level Shape Estimation, *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026, to appear. [arXiv][Website][video]

[3] Yue Hu, Junzhe Wu, **Ruihan Xu**, Hang Liu, Avery Xi, Henry X. Liu, Ram Vasudevan, Maani Ghaffari, Imaginative World Modeling With Scene Graphs Via VLM for Object Navigation, *IEEE Robotics and Automation Letters (RA-L)*, 2026, to appear. [arXiv]

[4] Joey Wilson*, **Ruihan Xu***, Yile Sun, Parker Ewen, Minghan Zhu, Kira Barton, Maani Ghaffari, LatentBKI: Open-Dictionary Continuous Mapping in Visual-Language Latent Spaces With Quantifiable Uncertainty, *IEEE Robotics and Automation Letters (RA-L)*, vol. 10, no. 4, pp. 3102–3109, 2025. [arXiv][code]

[5] **Ruihan Xu**, Anthony Opipari, Joshua Jin Zhi Mah, Stanley Lewis, Haoran Zhang, Hanzhe Guo, Odest Chadwicke Jenkins, Single-View 3D Reconstruction via SO(2)-Equivariant Gaussian Sculpting Networks, *RSS Workshop on Geometric and Algebraic Structure in Robot Learning*, 2024. [arXiv][video]

[6] Jana Pavlasek, Joshua Jin Zhi Mah, **Ruihan Xu**, Odest Chadwicke Jenkins, Fabio Ramos, Stein Variational Belief Propagation for Multi-Robot Coordination, *IEEE Robotics and Automation Letters (RA-L)*, vol. 9, no. 5, pp. 4194–4201, 2024. [paper][Website][video]

RESEARCH EXPERIENCE

SPARK Lab, Massachusetts Institute of Technology — **Cambridge, MA**
Graduate Research Assistant, advised by Prof. Luca Carlone — *Sep 2025 – Present*

- Designed Argos, which adapts geometric foundation models for generalizable scene change detection, and integrated it into Argos-SLAM for lifelong mapping; built a large-scale change-detection benchmark and deployed the system on an AgileX bimanual manipulator [1].
- Designed a LiDAR–camera fusion pipeline for real-time, far-field, category-agnostic object shape and pose estimation, enabling online outdoor 3D scene graphs with complete 3D mesh object shapes [2].
- Deployed scene graph updating and navigation planning on Boston Dynamics Spot robots, enabling multi-robot collaborative mapping and mission execution from natural-language commands via PDDL planning.
- Developed scene graph tokenization based on Segmented Eulerian Neighborhood Trails (SENT) and fine-tuned a 3B-parameter Qwen model with supervised fine-tuning (SFT) to predict scene graphs in unobserved space.

Computational Autonomy and Robotics Laboratory (CURLY), University of Michigan — **Ann Arbor, MI**
Research Assistant, advised by Prof. Maani Ghaffari — *May 2024 – Aug 2025*

- Designed the Latent Bayesian Kernel Inference algorithm for open-dictionary continuous mapping in vision-language latent space, with uncertainty quantification and open-dictionary query ability [4].
- Contributed a VLM-based scene graph prediction mechanism in a POMDP formulation, and designed an object detection verifier and a traversability-based navigation map for faster object-goal navigation [3].
- Extended LatentBKI to a latent Gaussian Splatting map, combining fast training with open-dictionary queries and uncertainty quantification.

Laboratory for PROGRESS (L4P), University of Michigan — **Ann Arbor, MI**
Research Assistant, advised by Prof. Chad Jenkins — *Sep 2022 – Jun 2024*

- Designed Gaussian Sculpting Networks and an Extended Chamfer Distance loss for SO(2)-equivariant single-view 3D reconstruction, achieving faster inference than the state of the art at comparable quality [5].
- Built the MBot Bridge message-passing system and an MPC-style controller to deploy Stein Variational Belief Propagation on real robots; ran real-world experiments with 3 MBots in 4 environments [6].
- Developed a grasping pipeline with Dex-Net in PyBullet and deployed it on a Fetch mobile manipulator.

TEACHING AND ACADEMIC SERVICE

Instructional Aide, UMich Robotics: ROB 330 (Localization & Mapping), ROB 102 (Intro to AI & Programming) — *Aug 2023 – Dec 2024*
Reviewer: ICRA, IROS, RA-L, RSS, ICLR, NeurIPS — *Present*

HONORS AND AWARDS

James B. Angell Scholar, University of Michigan (2024) • University Honors, University of Michigan (Dec 2022, Apr 2023, Dec 2023) • Phi Kappa Phi National Honor Society (2023) • Phi Beta Kappa O'Brien Book Award, top 15 freshmen, University of Rochester (2022) • Dean's List, University of Rochester (4 semesters, 2020–2022) • Davis UWC Scholarship (2020–2022)

TECHNICAL SKILLS

Programming: Python, C++/C, MATLAB, Docker, Conda
Deep Learning: PyTorch, OpenCV, vision-language models, geometric foundation models
Robotics: ROS, SLAM, 3D scene graphs, Gaussian Splatting, Gazebo, PyBullet, Meta AI Habitat